

MOHD MUZZAMMIL

Mechanical Engineering Student | CAD Designer | Embedded Systems Enthusiast

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EDUCATION

Muffakham Jah College of Engineering & Technology

2024 – 2028

B.E. Mechanical Engineering | Hyderabad, Telangana

TECHNICAL SKILLS

CAD & Design: SolidWorks (3D Modelling, Assembly, Technical Drawing)

Robotics: ESP32, Arduino, Autonomous Systems, Sensor Integration, Servo Motors, ROS Basics

Programming: Python, Arduino C/C++, Embedded Firmware (vibe code)

Fabrication: Manual Prototyping, Chassis Fabrication, Mechanical Systems

COMPETITIONS & TEAM EXPERIENCE

ABU Robocon 2026 — Mechanical Designer

2025 – Present

- Serving as mechanical designer for the college team's ABU Robocon 2026 competition robot.
- Contributing to mechanism design, CAD modelling, and structural design within the SolidWorks environment.

SAE Baja — Chassis Design

2025 – Present

- Currently designing the chassis for the team's SAE Baja all-terrain vehicle.
- Working on structural layout, load-bearing analysis, and CAD modelling to meet SAE Baja safety and rulebook requirements.

PROJECTS

Smart Agri Four Legged Bot (S.A.F.L.)

Nov 2024 – May 2026

MJCET R&D Cell Project | CAD & Assembly Design | onerealti.github.io/astro-safl

- Contributed as CAD & Assembly Design lead on a 5-member team building a semi-autonomous, four-legged wheeled agricultural rover for precision farming.
- Handled mechanical modelling, component alignment checks, and physical assembly layout integration in SolidWorks for the wheeled-leg chassis.
- Supported structural validation work that achieved a minimum safety factor of 2.04 under static, dynamic, and soil-probe-insertion loads via FEA.
- Team SAFL-B won 1st place at the MAKEFORHYDERABAD Make-a-thon (May 2026).

Companion Robot

2024 – Present

Self-Initiated Project | github.com/MohdMuzzammil-Igtm/Companion-Robot

- Built an AI-powered autonomous two-wheeled companion robot using a smartphone as the onboard "brain" instead of a Raspberry Pi, keeping the total build cost under ₹5,000.
- Integrated an ESP32 hardware controller with an Android phone running Python (Termux) for speech recognition (Whisper) and text-to-speech.
- Connected a locally-hosted Llama 3.1 model (via Ollama) for offline conversational AI, with YOLOv8 for real-time human detection and dual ultrasonic obstacle avoidance.
- Designed voice-command and gamepad control schemes, an animated TFT mood display, and a browser-based touch joystick controller.
- Won 2nd prize at the MJCET Project Expo.

Steam Engine Model

2022 (10th Grade)

Personal Build Project

- Designed and hand-built a working steam engine model from scratch, applying principles of thermodynamics, mechanical linkages, and energy conversion.
- Presented at school science fair, earning recognition for practical engineering skills.

References available upon request | Open to internships and collaborative engineering projects